FOODVERSE AI

Multimodal Food Intelligence Agent

PROJECT TESTING

1. Testing Objective

The purpose of this phase is to verify that the Foodverse AI backend starts, implemented API endpoints respond as expected, and invalid or unusual input is handled safely. Testing also helps identify integration issues before demonstration or deployment.

2. Testing Environment

- Windows Command Prompt or PowerShell and the project virtual environment (.venv).
- Python backend using FastAPI and Uvicorn.
- FastAPI Swagger UI, usually available at /docs while the local server is running.
- Recipe endpoint discussed during testing: POST /api/v1/recipes/generate.

3. Testing Approach

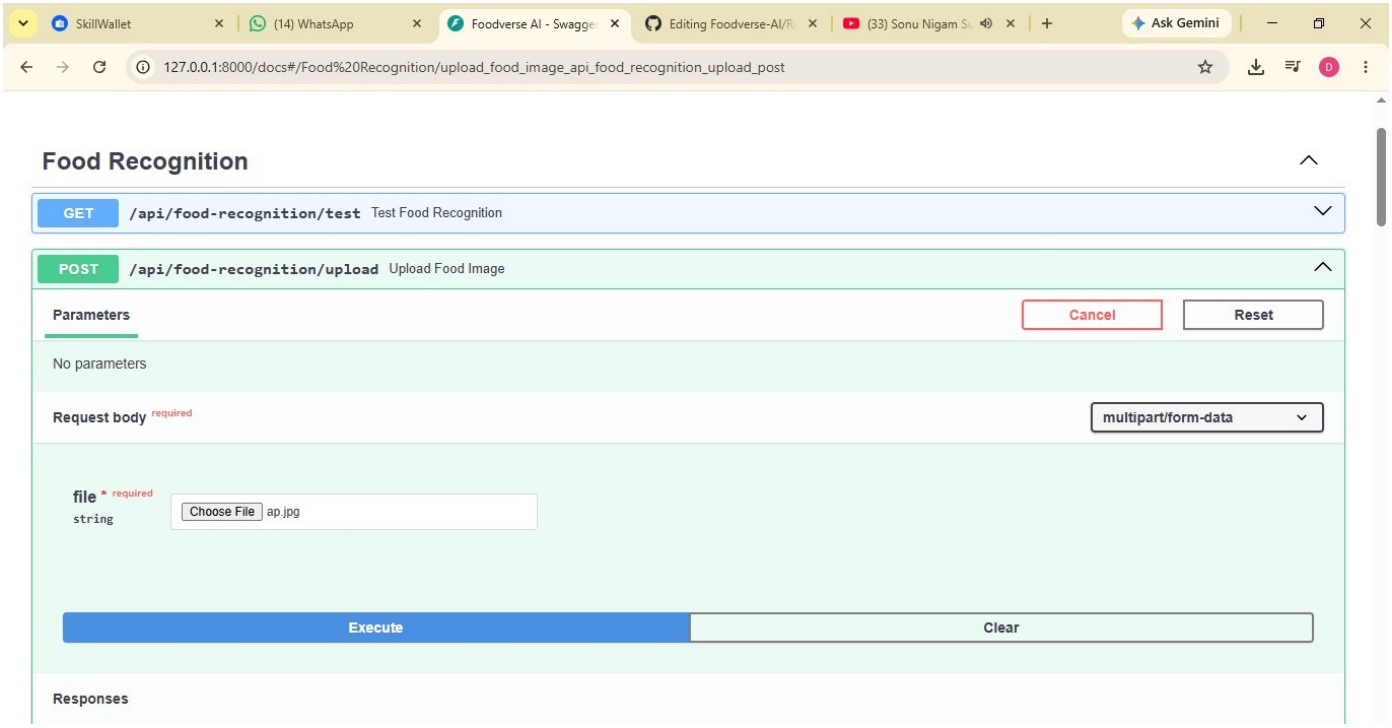
Testing was performed incrementally: start the backend, open the API documentation, submit requests, and inspect responses. Input cases included a normal recipe request and empty, whitespace-only, or unusual ingredient input. A successful HTTP status confirms API-level processing, not necessarily AI accuracy.

4. Test Cases

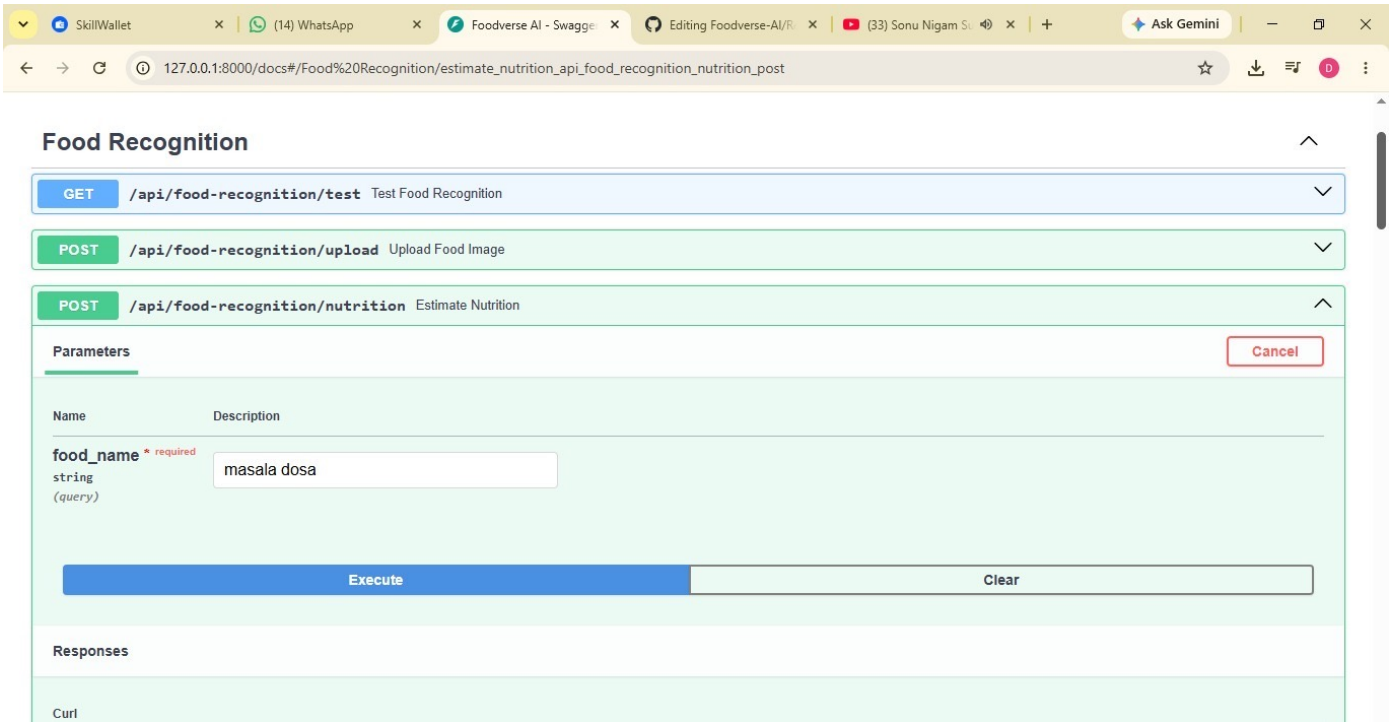
ID	Test	Expected result	Recorded status
TC-01	Start backend	Server starts without fatal error.	Reported successful.
TC-02	Open /docs	Swagger UI loads.	Used for endpoint testing.
TC-03	Valid recipe request	Supported fields accepted and response returned.	HTTP 200 reported for a test request.
TC-04	Empty ingredient input	Clear validation response; no crash.	Tested; exact response not recorded here.
TC-05	Whitespace-only input	Handled safely with clear response.	Tested during validation sequence.
TC-06	Unusual ingredients	No unhandled server failure.	Tested during validation sequence.
TC-07	Check FastAPI/Uvicorn	Package details displayed.	Package check completed.

Activity 2: Food Recognition and Nutrition Module Development

Activity 2.1: Food Image Detection and Classification

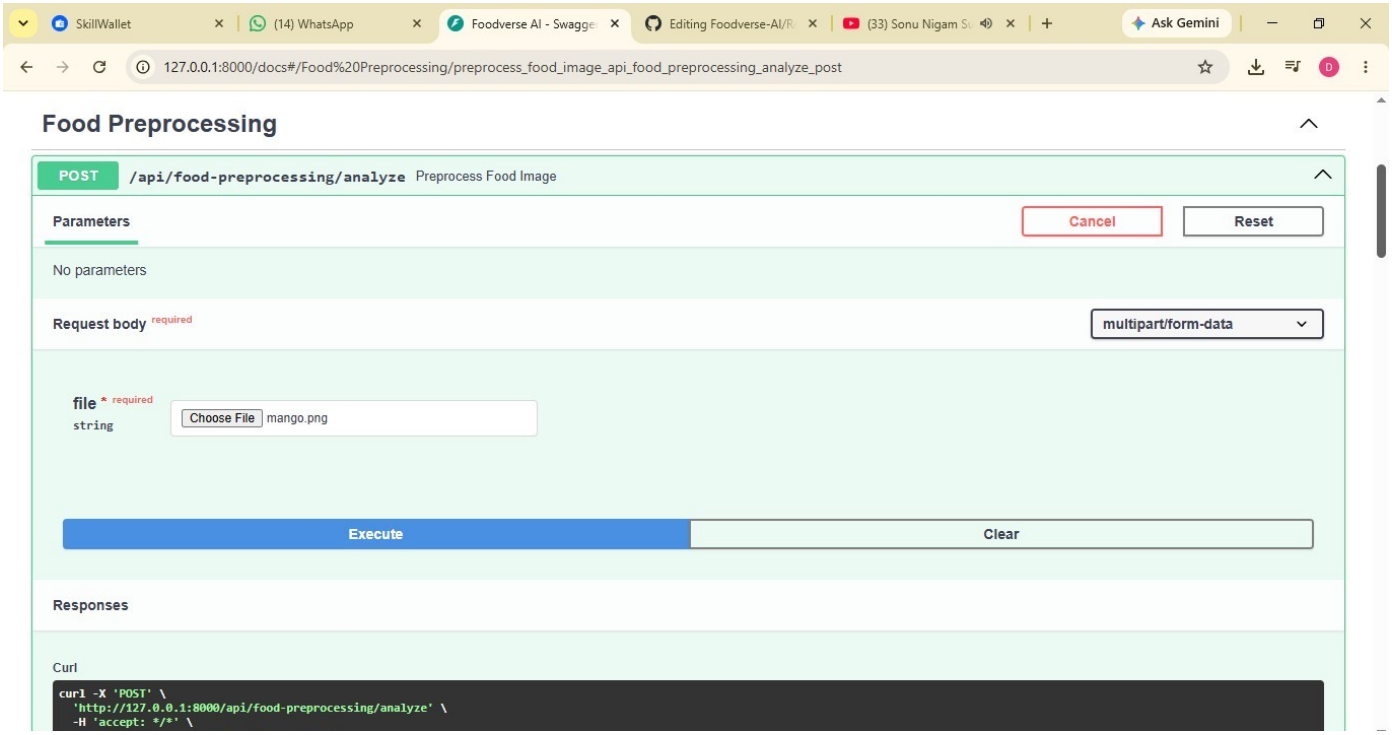


Activity 2.2: Nutrition Estimation and Health Scoring Logic

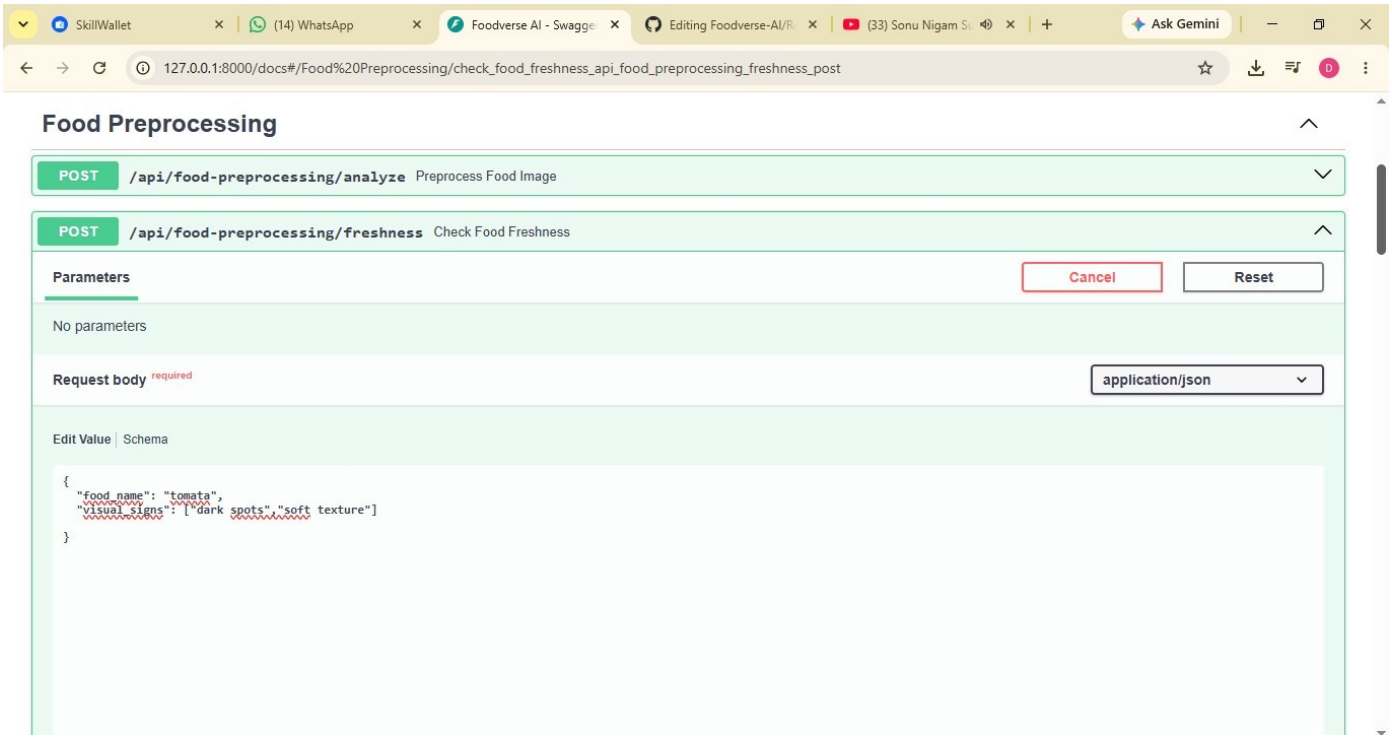


Activity 3: Food Freshness and Spoilage Detection Module

Activity 3.1: Food Image Processing and Feature Extraction

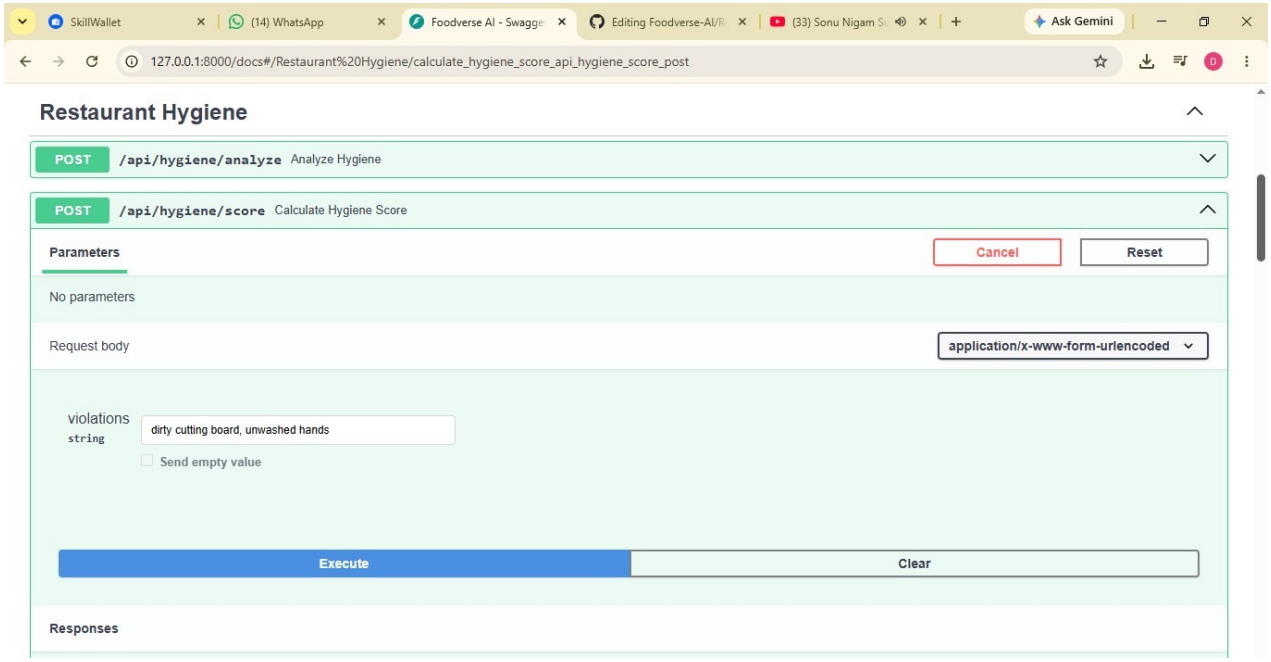


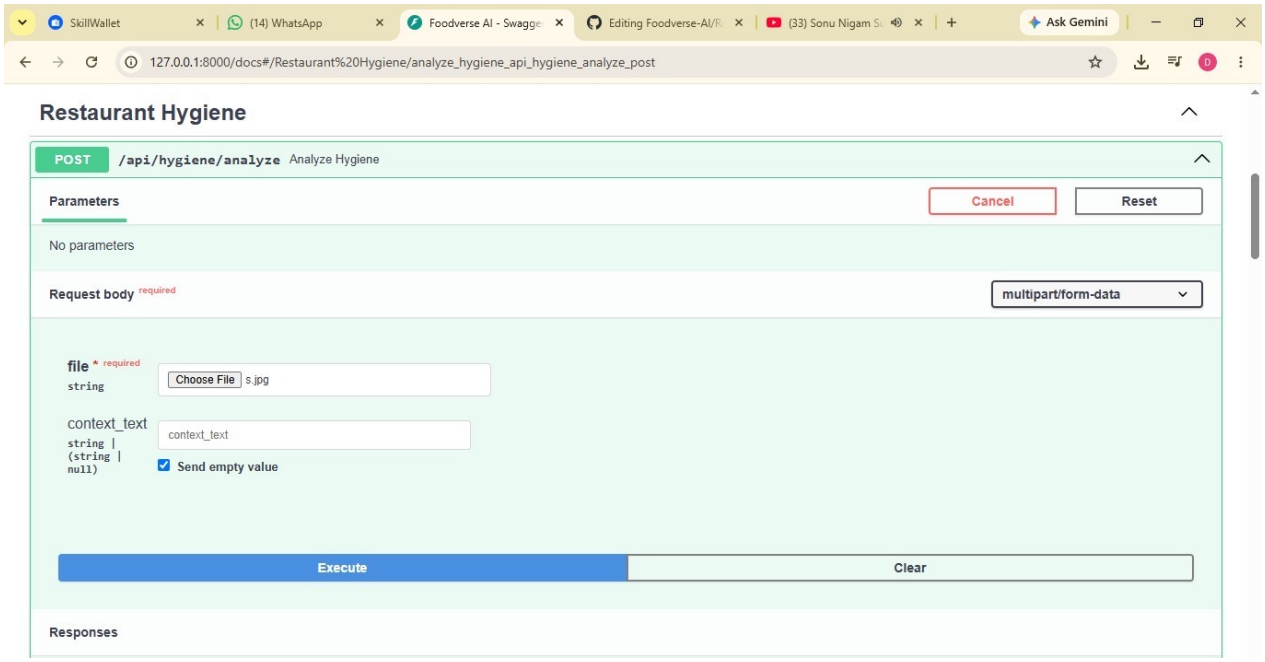
Activity 3.2: Freshness Classification and Safety Recommendation Logic



Activity 4: Restaurant Hygiene and Smart Kitchen Modules

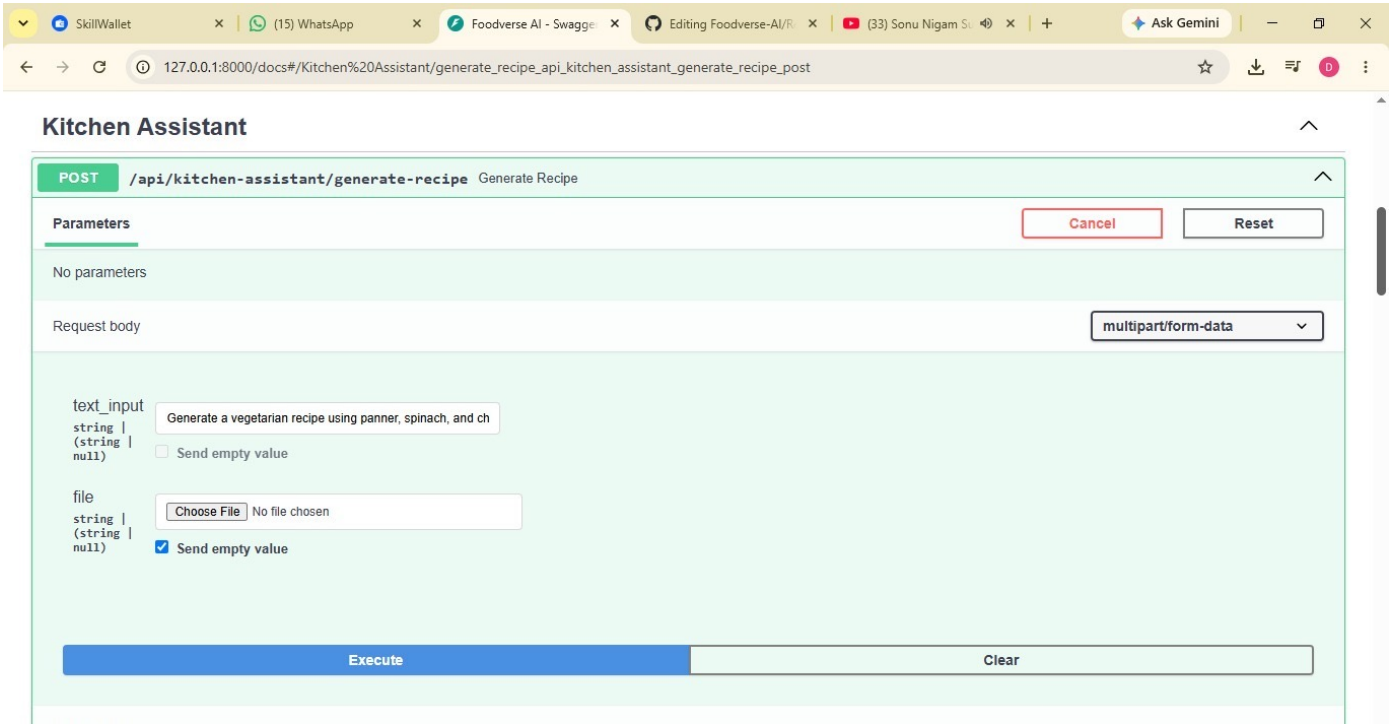
Activity 4.1: Restaurant Hygiene Detection and Compliance Scoring





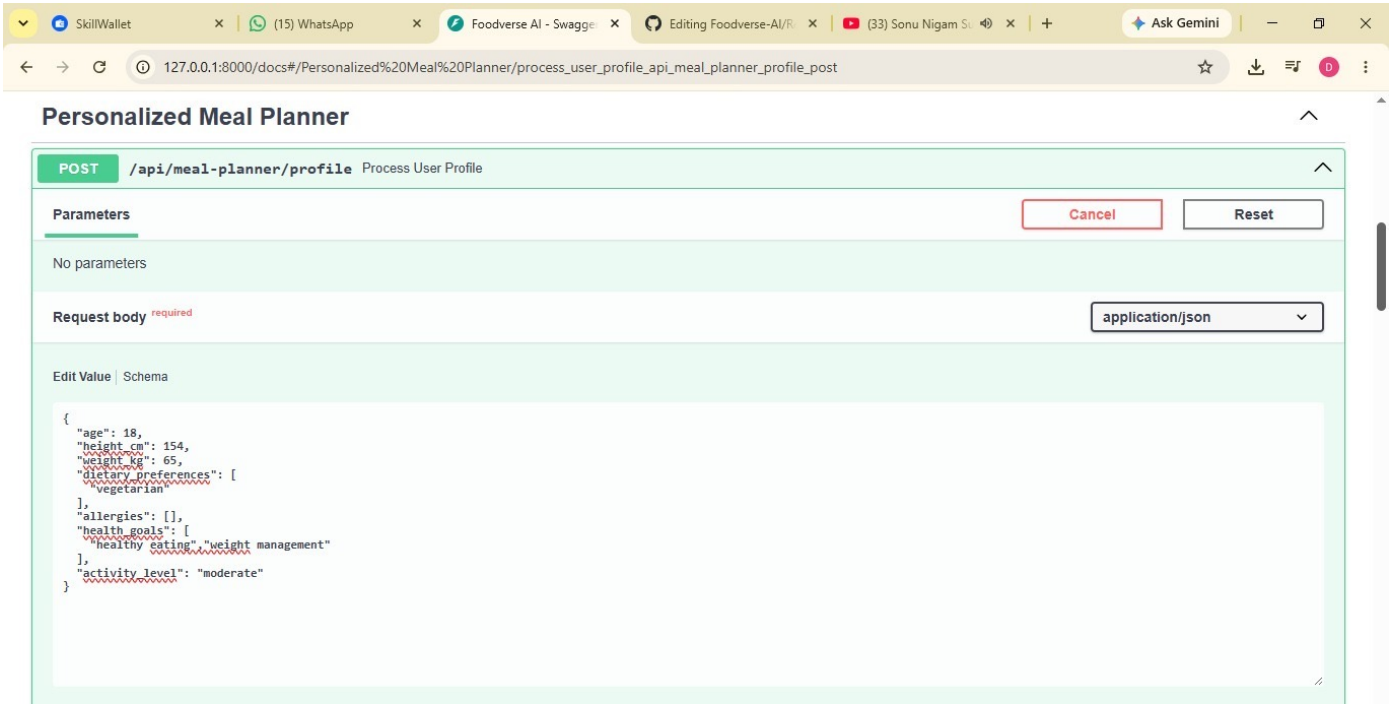
Activity 4: Restaurant Hygiene and Smart Kitchen Modules

Activity 4.2: Smart Kitchen Cooking Guidance and Recipe Generation Logic



Activity 5: Personalized Meal Planner Development

Activity 5.1: User Profile, Goal and Preference Processing



Activity 5: Personalized Meal Planner Development

Activity 5.2: Meal Plan Generation and Adaptive Recommendation Logic

